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Deterministic pose-spline dynamics estimator

Estimator structure
  pose-only continuous-time spline -> analytic derivatives
  fixed-spline gradient matching -> shared physical parameters and lag
  shooting nodes: none
  continuity constraints: none
  sensor/IMU channels in loss: none

Initial estimate
  source kind: multiple_shooting_result
  source path: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-
    estim/minimal/output/deterministic_multiple_shooting_multi/result.json
  source mass [kg]: 3.02232298411
  selected initial mass [kg]: 3.02232298411

Selected strict-ZOH parameters
  mass [kg]: 2.76395719961
  inertia [kg m^2]:
    [ 0.0727819009231, -0.00157574128676,  0.00017021215867]
    [-0.00157574128676,  0.145464959033,  0.000121049039743]
    [ 0.00017021215867,  0.000121049039743,  0.217282275919]
  principal moments [kg m^2]: [0.072747548907, 0.145498912684, 0.217282674284]
  inertia triangle margin [kg m^2]: 0.00096378730726
  CoG offset [m]: [-0.0028548317789, -0.000259983964135, -0.0162912081759]
  rotor force effectiveness: [1.00294882179, 1.03446692886, 0.972415930905, 0.991180047602]
  lag [s]: 0.048338405532
  joint dynamics loss: 0.383758618049
  soft-prior cost: 0.0456807044696

Smooth physical coordinates
  log_mass_scale                                       0.161568497783
  log_second_moment_cholesky_xx_scale                  0.405188852452
  log_second_moment_cholesky_yy_scale                  0.0568051583329
  log_second_moment_cholesky_zz_scale                  0.00166042384935
  normalized_second_moment_cholesky_yx_offset          0.0162839350461
  normalized_second_moment_cholesky_zx_offset         -0.00498864757033
  normalized_second_moment_cholesky_zy_offset         -0.00593920262519
  cog_offset_x_m                                      -0.0028548317789
  cog_offset_y_m                                      -0.000259983964135
  cog_offset_z_m                                      -0.0162912081759
  force_effectiveness_contrast_1                      -0.0218791332027
  force_effectiveness_contrast_2                       0.0378748429016
  force_effectiveness_contrast_3                       0.0102295826074

Spline and collocation
  collocation step [s]: 0.01

Bag failure_1
  selected spline knot spacing [s]: 0.1
  dynamics loss: 0.669190097396
  estimated forward position RMSE [m]: 23.1316402934
  nominal forward position RMSE [m]: 56.2552485683
  estimated improves nominal: True
  residual wrench mean: [ 0.088042  0.078711  0.058019 -0.0298   -0.009029 -0.00958 ]
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  residual wrench RMSE: [1.122817 0.737714 1.471983 0.128496 0.632377 0.427237]
  residual wrench trend/s: [ 0.105839  0.041043 -0.567822  0.003477 -0.017555  0.02326 ]
  residual wrench cumulative impulse: [ 0.541265  0.450173  0.330023 -0.177132 -0.054916
    -0.057299]

Bag failure_2
  selected spline knot spacing [s]: 0.2
  dynamics loss: 0.098327138702
  estimated forward position RMSE [m]: 51.3112593647
  nominal forward position RMSE [m]: 50.2776651111
  estimated improves nominal: False
  residual wrench mean: [ 0.070862 -0.362079 -0.019717  0.026933  0.024305  0.01447 ]
  residual wrench RMSE: [0.246384 0.447398 0.712196 0.04521  0.2421   0.08776 ]
  residual wrench trend/s: [-0.039577  0.030114 -0.281661 -0.009902  0.046324  0.00336 ]
  residual wrench cumulative impulse: [ 0.391955 -1.966245 -0.117245  0.146912  0.127858
    0.078193]

Smoothstep continuation
  width=0.5: delay=0.0476858722s, cost=0.429349362069, nfev=16
  width=0.2: delay=0.0484578455s, cost=0.429467756736, nfev=10
  width=0.05: delay=0.0483384055s, cost=0.429437416086, nfev=4

Strict-ZOH polish
  delay=0.0443384055s: screening=0.430320116564, refined=nan, selected=False
  delay=0.0453384055s: screening=0.430384233902, refined=nan, selected=False
  delay=0.0463384055s: screening=0.430296654816, refined=nan, selected=False
  delay=0.0473384055s: screening=0.43208906756, refined=nan, selected=False
  delay=0.0483384055s: screening=0.429439324806, refined=0.429439322519, selected=True
  delay=0.0493384055s: screening=0.429862691903, refined=0.429861764707, selected=False
  delay=0.0503384055s: screening=0.429868798721, refined=0.429867832441, selected=False
  delay=0.0513384055s: screening=0.429975216206, refined=nan, selected=False
  delay=0.0523384055s: screening=0.432475568738, refined=nan, selected=False
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